NUMBER SYSTEM

NUMBER SYSTEM

- Modern computers communicate and operate with binary numbers 0 and 1.Human beings used decimal number system where as system uses number system.

- Consider an example

  If we use decimal number 4, we represent Binary number as 100.

  Consider another example

  If we use decimal number 21, we represent Binary number as 10101.

- In above example we require 2 digits for decimal number and 5 digits for Binary number. Therefore we can say that for when decimal quantities are in the binary form it require more digits.
- So for large decimal numbers it requires large binary digits, so it should be some

  inconvenient to user.

NUMBER SYSTEM

Number System can be represented in 4 ways

1) Decimal Number System

2) Binary Number System

3) Octal Number System

4) Hexa decimal Number System

NUMBER SYSTEMS

SYSTEM	BASE	SYMBOLS	USED BY HUMANS?	USED BY COMPUTERS?
Decimal	10	0,1,2-------9	Yes	No
Binary	2	0,1	No	Yes
Octal	8	0-7	No	No
HEXA-DECIMAL	16	0-15	No	No

DECIMAL NUMBER SYSTEM

- In decimal number system we can express any decimal number in units i.e, tens, hundreds, thousands .....

  Example: 5679

- We can represent the above example as 5000 + 600 + 70 + 9

- The position of the digit with reference to the decimal point determines the value/weight.

- The sum of all digits multiplied by their weights givens the total number being represented.

- The left most digit, which has the greatest weight is called the Most Significant Bit (MSB).

- The Rightmost digit, which has the least weight is called the Least Significant Bit (LSB).

BINARY NUMBER SYSTEM

- A binary number is a number expressed in the binary numeral system or base-2 numeral system which represents numeric values using two different symbols: typically 0 (zero) and 1 (one). The base-2system is a positional notation with a radix of 2.
- Applications of Binary numbers:

1. The binary number system used with computers to recognize only two states of operation

   i.e. on(1)/off(0)

2. Binary numbers are easily transformed into electrical signals.
3. Binary numbers can be effectively used to encrypt messages.
4. Binary number system used in statistical investigations, prove mathematical theorems and solve puzzles.

OCTAL NUMBER SYSTEM

- The octal numeral system, or **oct** for short, is the base-8 number system, and uses

  the digits 0 to 7.

- Octal numerals can be made from binary numerals by grouping consecutive binary

  digits into groups of three (starting from the right).

  For example, the binary representation for decimal 74 is 1001010.

Octal Symbol	Binary equivalent
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

HEXA – DECIMAL NUMBER SYSTEM

- In mathematics and computing, hexadecimal (also base **16**, or hex) is a positional numeral system with a radix, or base, of **16**. It uses **sixteen** distinct symbols, most often the symbols 0–9 to represent values zero to nine, and A, B, C, D, E, F (or alternatively a, b, c, d, e, f) to represent values ten to fifteen.

Binary	Decimal	Hexadecimal
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	10	A
1011	11	B
1100	12	C
1101	13	D
1110	14	E
1111	15	F

DECIMAL NUMBER SYSTEM

- Decimal Number system conversions
  - ✔ **Decimal to binary conversion**
  - ✔ **Decimal to Octal conversion**
  - ✔ **Decimal to Hexa Decimal**

DECIMAL TO BINARY CONVERSION

- Divide the number by the 'Base' (=2)
- Take the remainder (either 0 or 1) as a coefficient
- Take the quotient and repeat the division

Example: $(14)_{10}$

	Quotient	Remainder	Coefficient
14 / 2 =	7	0	$a_0 = 0$
7 / 2 =	3	1	$a_1 = 1$
3 / 2 =	1	1	$a_2 = 1$
1 / 2 =	0	1	$a_3 = 1$

Answer: $(14)_{10} = (a_3\, a_2\, a_1\, a_0)_2 = (1110)_2$

MSB LSB

Decimal to Binary Conversion

Example : Convert the given decimal number 24.6 into binary

Integer part:24

Fractional part:0.6

Step-1 (one approach)

Divide 24 by 2

2) 24 (12
-24
0

Divide 12 by 2

2) 12 (6
-12
0

Divide 6 by 2

2) 6 (3
-6
0

Divide 3 by 2

2) 3 (1
-2
1

Divide 1 by 2

2) 1 (0
-0
1

R	
0	LSD
0	
0	≡
1	
1	MSD

Step-1 (another approach)

	Q	R
2	24	0
2	12	0
2	6	0
2	3	1
2	1	1
2	0	

LSD

MSD

Step:2

Find the equivalent Binary number for fractional part

Fraction		Radix		Result			Recorded carries	
0.6	*	2	=	1.2	=	0.2	with a carry	1
0.2	*	2	=	0.4	=	0.4	with a carry	0
0.4	*	2	=	0.8	=	0.8	with a carry	0
0.8	*	2	=	1.6	=	0.6	with a carry	1

MSD

LSD

Therefore $(24.6)_{10} = (11000.1001)_2$

DECIMAL TO OCTAL CONVERSION

Example: $(160)_{10}$

		Quotient	Remainder	Coefficient
160	/ 8 =	20	0	$a_0 = 0$
20	/ 8 =	2	4	$a_1 = 4$
2	/ 8 =	0	2	$a_2 = 2$

Answer: $(160)_{10} = (a_2\, a_1\, a_0)_8 = (240)_8$

Decimal to Octal Conversion

Example : Convert the given decimal number 35.45 into octal

Integer part:35

Fractional part:0.45

Step-1

Divide 35 by 8

```
8)  35 (4
    32
     3
```

Divide 4 by 8

```
8)  4  (4
    0
    4
```

R	
3	LSD
4	MSD

≡

	Q	R
8	35	
	4	3

LSD

MSD

Step:2

Find the equivalent octal number for fractional part

Fraction		Radix		Result			Recorded carries	
0.45	*	8	=	3.6	=	0.6	with a carry	3
0.6	*	8	=	4.8	=	0.8	with a carry	4
0.8	*	8	=	6.4	=	0.4	with a carry	6
0.4	*	8	=	3.2	=	0.2	with a carry	3

MSD

LSD

Therefore $(35.45)_{10} = (43.3463)_8$

DECIMAL TO HEXADECIMAL CONVERSION

Convert $(40)_{10}$ to its hexadecimal equivalent:

40 / 16 = 2 R 8

2 / 16 = 0 R 2

$\boxed{(28)_{16}}$

DECIMAL TO HEXADECIMAL CONVERSION

Example :
Convert the given decimal number 22.64 into hexadecimal in this example the number contains fractional part so the fractional part is multiplied with radix r

Integer part:22

Divide 22 by 16

```
16) 22 (1
    16
     6
```

Divide 6 by 16

```
16) 1 (1
    0
    1
```

R	
6	LSD
1	MSD

≡

fractional part:0.64

	Q	R	
16	22	6	LSD
16	1	1	
	0		MSD

Step:2

Find the equivalent hexadecimal number for fractional part

Fraction		Radix		Result			Recorded carries			
0.64	*	16	=	10.24	=	0.24	with a carry	10	=	A
0.24	*	16	=	3.84	=	0.84	with a carry	3	=	3
0.84	*	16	=	13.44	=	0.44	with a carry	13	=	D
0.44	*	16	=	7.04	=	0.04	with a carry	7	=	7

MSD

LSD

Therefore $(22.64)_{10} = (16.A3D7)_{16}$